

Hand Gesture Classification Using Two-Channel sEMG for Rehabilitation

Applications: A Low-Cost Approach with Edge Impulse

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Abstract— This project presents a gesture classification system based on surface electromyography (sEMG). Signals are acquired from two electrode channels placed on the forearm (over the finger flexor and extensor muscles). The goal is to integrate this system into the development of myoelectric prostheses that more accurately recognize the movements the user intends to perform. A neural network was trained in Edge Impulse Studio to distinguish between four movements: rest, closed fist, power grip, and index finger extension. The training data were obtained from the Ninapro DB2 dataset. The final model achieved an accuracy of 57.0% during validation and 51.4% with a completely independent subject (cross-subject testing); this configuration is presented as the primary result of this work.

Keywords— sEMG, clasificación de gestos, Ninapro DB2, generalización entre sujetos, Edge Impulse, rehabilitation, Streamlit.

I. Introduction

Surface electromyography (sEMG) has become a widely used non-invasive technique for capturing muscle activation patterns, with applications ranging from prosthetic control to rehabilitation monitoring [1]. Wearable EMG-based systems are increasingly proposed as tools to extend rehabilitation monitoring beyond the clinic, enabling therapists to track patient progress and adjust treatment remotely [3].

A well-known and persistent challenge in EMG-based gesture recognition is generalization across subjects: models trained on one group of individuals often perform substantially worse on a new, unseen subject, due to anatomical variability, electrode placement differences, and skin/tissue impedance differences [2]. This gap is especially pronounced in systems that use a reduced number of EMG channels, as opposed to the 8–12 channel arrays typically used in high-performance research systems.

This project explores whether a lightweight, embeddable model trained using Edge Impulse Studio in conjunction with a minimal 2-channel sEMG setup (flexor and extensor digitorum superficialis) can achieve usable gesture classification while being open about its cross-subject limitations, an aspect of sEMG research that is frequently underreported.

II. Problem Statement

Strong classification accuracy is achieved by high-channel-count EMG systems, but they are more expensive, need more electrodes, and have more complicated signal chains, making them unsuitable for low-resource environments and quick prototyping. In the meantime, cross-subject generalization is usually deteriorated in systems that limit the number of channels for cost and simplicity, and crucially, this degradation is rarely assessed explicitly.

This project addresses the following question: using only 2 sEMG channels and a compact neural network suitable for edge deployment, what gesture classification performance can realistically be achieved, both within a pool of known subjects (validation) and on a completely unseen subject (cross-subject test), and what does the accuracy gap between these two scenarios reveal about the design's limitations?

III. Proposed Solution

A gesture classification pipeline was built around the Ninapro DB2 dataset [1], restricted to channels 9 and 10 (over flexor and extensor digitorum superficialis) to emulate a minimal, low-cost 2-channel acquisition setup. Four gesture classes were defined: rest, closed fist, medium wrap, and index finger extension. A dense neural network was trained and exported via Edge Impulse Studio as a TensorFlow Lite model, small enough for on-device or lightweight web deployment. A Streamlit application was built to load real sEMG recordings (BITalino/OpenSignals format), visualize them, segment them by expected gesture, and run the trained model for offline evaluation.

IV. Methodology

A. Data Preparation

Data was drawn from Ninapro DB2 [1], a benchmark sEMG dataset comprising 12-channel recordings at 2000 Hz from 40 intact subjects performing hand, wrist, and grasp movements. Only channels 9 and 10 (flexor and extensor digitorum superficialis) were used, discarding the remaining 8-channel forearm array to emulate a minimal-channel acquisition setup.

The corrected post-hoc label (restimulus) was used instead of the raw stimulus cue, to avoid including onset/offset transition samples within movement segments. Each movement repetition was trimmed by 100 ms on each end to further reduce onset/offset ambiguity. Rest periods, which are much longer than movement periods in the raw dataset, were capped at a maximum of 6 segments per source file, each limited to 3 seconds, to avoid overrepresenting rest in the training data.

Per-subject z-score normalization was a crucial preprocessing step. For each subject and channel, the mean and standard deviation were calculated from that subject's own rest-period samples and used to normalize all of that subject's samples (gesture and rest). This was inspired by an empirical finding: the standard deviation of the rest-period signal changed by up to $6\times$ between participants in the absence of normalization, suggesting that the raw amplitude scale carries significant inter-subject variability unrelated to gesture content.

The pipeline extracted raw data from .mat files in Python, then performed windowing (200 ms windows, 100 ms step) and artifact filtering (based on a peak-to-RMS ratio heuristic) in Google Colab, before uploading samples to Edge Impulse via its SDK.

B. Model Architecture

A total of 14 input features (7 statistics $\times$ 2 channels: flexor and extensor digitorum superficialis) were extracted using Edge Impulse's Flatten DSP block, which computed seven statistics per channel over each 100 ms window: mean, minimum, maximum, RMS, standard deviation, skewness, and kurtosis. The deployed TensorFlow Lite model, whose input tensor has form, was used to explicitly verify this [1, 14].

The classifier is a dense (totally connected) neural network with a 4-class softmax output layer after two hidden layers of 20 and 10 neurons, respectively. Instead of depending only on a random train/validation split within the same subject pool, the subjects were divided so that five were used for training/validation and one was kept out entirely for cross-subject testing. This allowed for an explicit measurement of generalization to an unseen individual.

C. Model Evaluation

Two methods were used to assess the model's performance: (1) validation accuracy inside the training subject pool, and (2) cross-subject test accuracy on the fully held-out subject. Since the deployed system uses this setup, per-subject z-score normalization was implemented in both situations.

Scenario	Accuracy	F1 (rest)	F1 (gestures, range)
Validation	57.0%	0.77	0.37 – 0.59
Test (held-out subject)	51.4%	0.94	0.15 – 0.42

Table 1.— Validation vs. cross-subject test performance

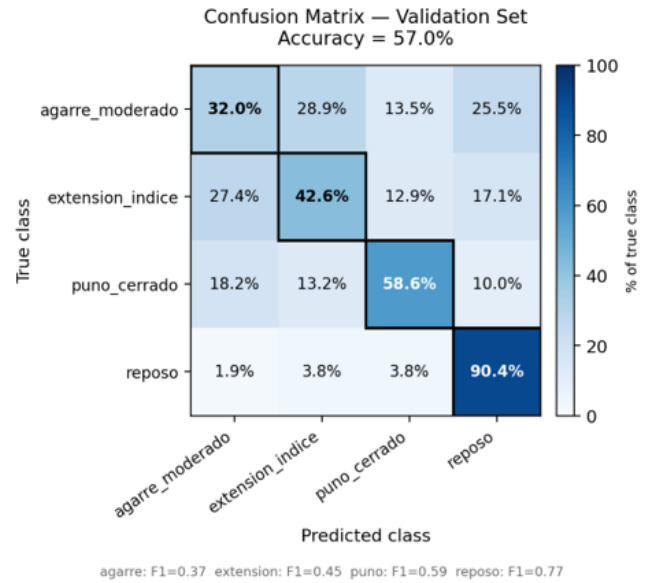


Figure 1. Confusion matrix — validation set.

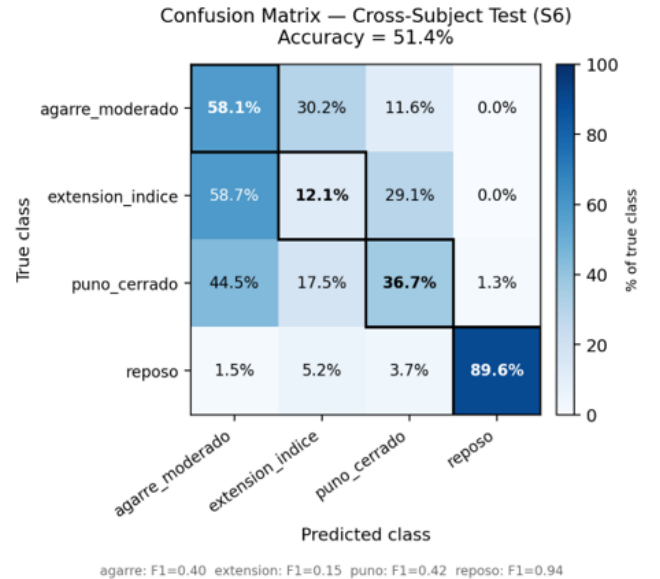


Figure 2. Confusion matrix — cross-subject test on isolated subject S6.

A recurring pattern appeared in both situations: rest was the class that was easiest to generalize because it simply corresponds to the presence or absence of muscle contraction, while differentiating between different grasp types across subjects proved to be much more difficult. This is likely because there are only two EMG channels available, which provide less spatial information about which specific muscles are activated for a given grasp type.

D. Confidence Analysis

Beyond the validation/test accuracy comparison above, per-window prediction confidence and probability distributions were inspected to characterize model behavior. In cross-subject testing without normalization, approximately 98% of movement-gesture predictions fell below the confidence threshold and were flagged as "uncertain", indicating the model was effectively unable to generalize at all to a new subject without normalization — a stronger and more informative finding than the raw accuracy number alone.

V. App Implementation

To facilitate the offline assessment of the trained model against actual sEMG recordings, a Streamlit web application was created. The program accepts both pre-processed CSV files and raw BITalino/OpenSignals.txt exports, which convert ADC data to millivolts using the manufacturer's EMG conversion algorithm. It offers manual segmentation of a continuous recording into labeled movement/rest segments, raw signal viewing for both channels, and traditional time-domain feature inspection (RMS, MAV, STD, waveform length, zero-crossings) per window. The Edge-Impulse-exported TensorFlow Lite model is used for live inference; converted signals, features, and predictions can be exported as CSV. Per-window prediction, confidence, and a confusion matrix comparing predicted versus expected gesture across all labeled segments of a recording are all included.

The application fully integrates live model inference: a.tflite model can be uploaded via the sidebar, and windowed inference is performed over the duration of the recording, reporting the dominant predicted gesture, average confidence, accuracy, and a complete confusion matrix against the expected labels for recordings with defined segments.

VI. Results

As summarized in Table 1, the resulting model achieved 57.0% accuracy on validation and 51.4% accuracy on the held-out cross-subject test (subject S6) using the per-subject z-score normalization setup used for this system. In both cases, rest was the most dependable class ($F1 = 0.77$ on

validation, 0.94 on cross-subject test). The three movement gestures showed greater confusion among themselves, especially closed fist, which was often misclassified as medium wrap (33.6–44.5% depending on evaluation set). These are the operating results reported for this work.

At the same time, the residual difference between validation (57.0%) and cross-subject test (51.4%) accuracy, both significantly higher than chance (25% for four classes), is relatively small, indicating that per-subject z-score normalization significantly reduced, but did not completely eliminate, the generalization gap commonly seen in cross-subject sEMG classification [2].

The trained model was assessed using the Streamlit program on a brief real-hardware recording made with a BITalino device (1000 Hz) that included three manually labeled segments: closed fist, medium wrap, and index-finger pointing. This served as a first qualitative check. The application used interpolation to resample the incoming data prior to inference because the deployed model requires 2000 Hz input. Two of the three segments—closed fist and medium wrap—were correctly classified using the subject's own rest period as the normalization baseline, but the pointing gesture was incorrectly labeled as closed fist. With small margins (2.8–7.3%) between the top two projected classes, prediction confidence was poor in all three segments (29.4–42.6%) and always fell below the application's uncertainty threshold. This suggests that, beyond the cross-subject generalization gap already observed on Ninapro DB2, an additional domain shift is likely present when moving from the original acquisition hardware (Delsys Trigno, 2000 Hz) to a different low-cost device (BITalino, 1000 Hz, resampled) with different electrode placement and skin contact conditions. Given the very small sample size ($n = 3$ segments, one subject), this result is reported as an exploratory finding rather than a validated accuracy figure; a systematically labeled BITalino dataset across multiple subjects is identified as necessary future work to properly quantify real-hardware performance.



Figure 3. Preliminary check through the Streamlit application.

VII. Conclusion

This work demonstrates that per-subject z-score normalization is a low-cost, high-impact intervention for

improving generalization to unseen subjects, and that a minimal 2-channel sEMG gesture classification system trained with a compact Edge-Impulse-deployable model can achieve meaningful, though not clinical-grade, cross-subject gesture recognition. In keeping with best practices in the sEMG generalization literature [2], the ongoing validation/test accuracy gap is stated openly rather than minimized.

This study's limitations include the limited sample size (6 of 40) compared to the entire Ninapro DB2 pool, the limitation to 2 of the 12 available EMG channels, and the lack of clinical validation with patient groups. Scaling to a larger subject pool, investigating spectral/frequency-domain features in addition to time-domain statistics, assessing additional channel configurations, and further validating the live-inference pipeline in the Streamlit application against actual BITalino recordings across numerous new users are all examples of future work.

References

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